US 20230338546A1

(19) **United States**

(12) **Patent Application Publication**

**WILSON et al.**

(10) **Pub. No.: US 2023/0338546 A1**

(43) **Pub. Date: Oct. 26, 2023**

(54) **ORAL FORMULATIONS OF KAPPA OPIOID RECEPTOR AGONISTS**


(71) Applicant: **Cara Therapeutics, Inc.**, Stamford, CT (US)

(72) Inventors: **Bryan R. WILSON**, Brewster, NY (US); **Stephen J. O'CONNOR**, Guilford, CT (US)



(21) Appl. No.: **18/211,787**

(22) Filed: **Jun. 20, 2023**

**Related U.S. Application Data**

(63) Continuation of application No. 17/242,678, filed on Apr. 28, 2021, now Pat. No. 11,684,674, which is a continuation of application No. 16/570,333, filed on Sep. 13, 2019, now Pat. No. 11,033,629.

(60) Provisional application No. 62/731,802, filed on Sep. 14, 2018.

**Publication Classification**

(51) **Int. Cl.**

| | |
|---|---|
| *A61K 47/14* | (2006.01) |
| *A61K 38/07* | (2006.01) |
| *A61K 47/26* | (2006.01) |
| *A61K 31/485* | (2006.01) |
| *A61K 47/18* | (2006.01) |
| *A61K 9/107* | (2006.01) |
| *A61K 47/64* | (2006.01) |
| *A61K 47/54* | (2006.01) |
| *A61K 31/40* | (2006.01) |
| *A61K 9/48* | (2006.01) |
| *A61K 38/08* | (2006.01) |

(52) **U.S. Cl.**

CPC .............. ***A61K 47/545*** (2017.08); ***A61K 9/107*** (2013.01); ***A61K 9/4891*** (2013.01); ***A61K 31/40*** (2013.01); ***A61K 31/485*** (2013.01); ***A61K 38/07*** (2013.01); ***A61K 38/08*** (2013.01); ***A61K 47/14*** (2013.01); ***A61K 47/183*** (2013.01); ***A61K 47/186*** (2013.01); ***A61K 47/26*** (2013.01); ***A61K 47/64*** (2017.08)


(57) **ABSTRACT**


The invention provides formulations for oral delivery of a therapeutic agent wherein the formulation comprises a kappa opioid receptor agonist and an absorption enhancer, the absorption enhancer includes a medium chain fatty acid or a salt of a medium chain fatty acid; and a medium chain fatty acid glyceride. The kappa opioid receptor agonist may be embedded in an oligosaccharide, such as trehalose. Also provided are capsules containing the oral formulations of the kappa opioid receptor agonists and the absorption enhancer of the invention and methods use of these formulations for the prophylaxis and treatment of variety of kappa opioid receptor-associated diseases and conditions such as pain, pruritus and inflammation; the method comprising administering to the mammal the formulation comprising the kappa opioid receptor agonist and an absorption enhancer.

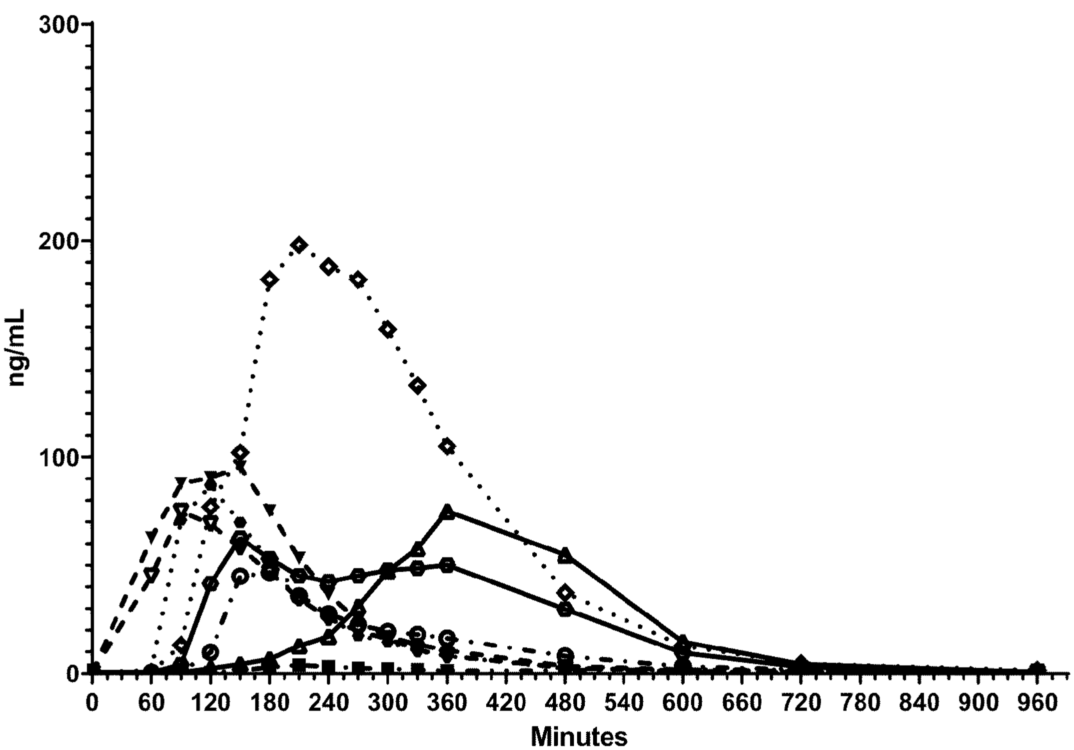
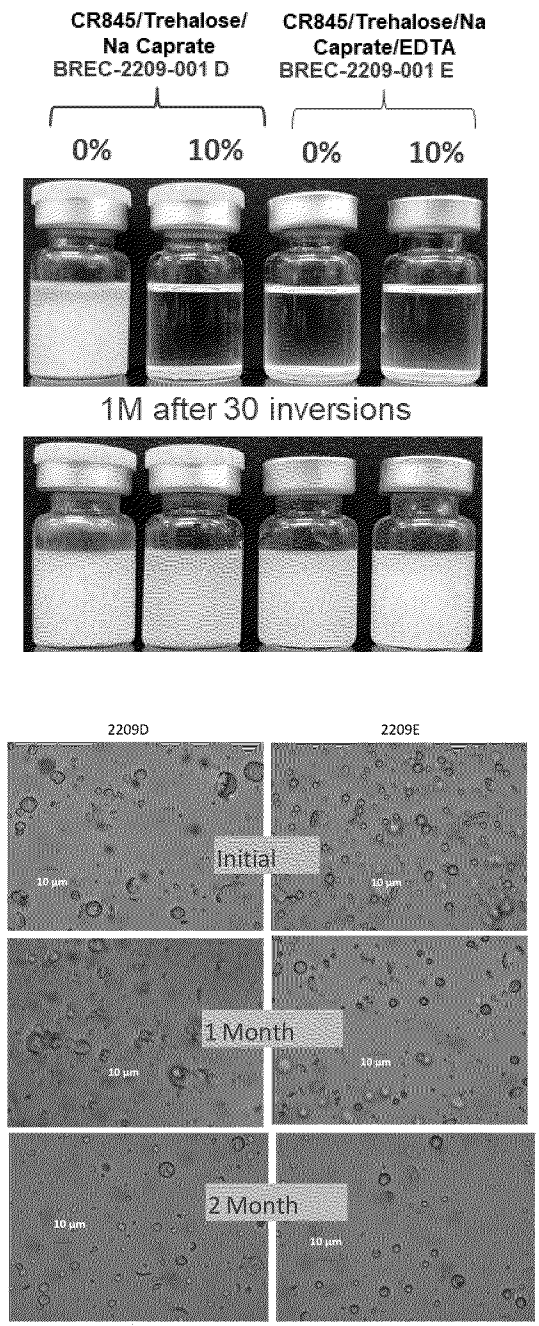
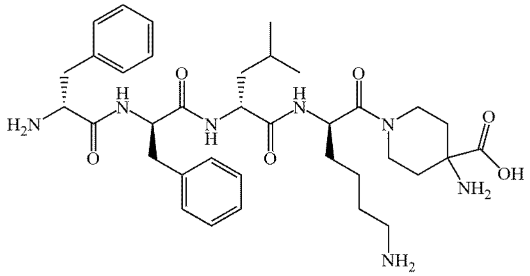



Fig. 9


CR845/Trehalose/
Na Caprate
BREC-2209-001 D
CR845/Trehalose/Na
Caprate/EDTA
BREC-2209-001 E
0%
10%
0%
10%
1M after 30 inversions
2209D
2209E
Initial
10 µm
1 Month
2 Month

Formula II



D-Phe-D-Phe-D-Leu-D-Lys-[ω(4-aminopiperidine-4-carboxylic acid)]-OH.

**[0013]** In still another embodiment, the oral formulation of the invention includes a peptide amide kappa opioid receptor agonist that includes one or more D-amino acids, and an absorption enhancer; wherein the synthetic peptide amide including one or more D-amino acids is CR845 and the absorption enhancer includes a medium chain fatty acid or a pharmaceutically acceptable salt of a medium chain fatty acid, and a medium chain fatty acid glyceride.

**[0014]** The present invention further provides methods of use of a formulation for oral delivery of a therapeutic agent that includes a kappa opioid receptor agonist and an absorption enhancer for the prophylaxis or treatment of kappa opioid receptor-associated diseases and conditions in a human patient or other mammal, the method comprising administering to the patient or the mammal the oral formulation of the invention. In one embodiment, the formulation of the invention includes a medium chain fatty acid or a salt of a medium chain fatty acid, and a medium chain fatty acid glyceride as absorption enhancers suitable for optimizing uptake of the kappa opioid receptor agonist from the gastrointestinal system and thereby enhancing its biological activity.

## BRIEF DESCRIPTION OF THE FIGURES

**[0015]** FIG. **1**: Bioavailability of CR845 post oral administration to canines (n=8) of a formulation containing (1) 1.6 mg spray dried CR845.HCl in 90% Miglyol® 812, 10% sodium caprate; (2) 1.6 mg crystallized CR845.HCl in 90% Miglyol® 812, 10% sodium caprate; (3) 1.6 mg spray dried CR845.HCl in 90% Miglyol® 812, 10% capric acid; (4) 1.6 mg spray dried CR845.HCl in 70% Miglyol® 812, 30% capric acid;.

**[0016]** FIG. **2**: Dissolution profile of formulation 1 containing 1.6 mg spray dried CR845.HCl in 90% Miglyol® 812, 10% sodium caprate on a scale of 0-300 mins.

**[0017]** FIG. **3**: Dissolution profile of formulation 2 containing 1.6 mg spray dried CR845.HCl in 90% Miglyol® 812, 10% capric acid on a scale of 0-300 mins.

**[0018]** FIG. **4**: Dissolution profile of formulation 3 containing 1.6 mg spray dried CR845.HCl in 70% Miglyol® 812, 30% capric acid on a scale of 0-300 mins.

**[0019]** FIG. **5**: Dissolution profile of formulation 4 containing 1.6 mg crystallized CR845.HCl in 90% Miglyol® 812, 10% sodium caprate on a scale of 0-300 mins.

**[0020]** FIG. **6**: Plasma concentration profile of formulation 5 containing 4.0 mg CR845.HCl, 20% capric acid in Miglyol® 812.

**[0021]** FIG. **7**: Plasma concentration profile of formulation 6 containing 4.0 mg CR845.HCl, 10% capric acid in Miglyol® 812.

**[0022]** FIG. **8**: Plasma concentration profile of formulation 7 containing 4.0 mg CR854, 5% capric acid in Miglyol® 812.

**[0023]** FIG. **9**: Suspensions of CR845.HCl with 0% or 10% sodium caprate; or 0% or 10% sodium caprate plus EDTA after standing and 30 inversions after I month storage at 40° C. Photomicrographs show consistent particle size after 1 month or 2 months storage at 40° C.

**[0024]** FIG. **10**: Plasma concentration profile of formulation 8 containing 2.0 mg spray dried CR845.HCl (21% w/w), trehalose (70% w/w), 9% (w/w) EDTA, suspended in 10% capric acid in Miglyol® 812.

**[0025]** FIG. **11**: Plasma concentration profile of formulation 9 containing 2.0 mg spray dried CR845.HCl (19% w/w), trehalose (62% w/w), sodium caprate (14% w/w), EDTA (5% w/w), in Miglyol® 812.

**[0026]** FIG. **12**: Plasma concentration profile of formulation 10 containing 2.0 mg spray dried CR845.HCl (18% w/w), trehalose (60% w/w), sodium caprate (13% w/w), EDTA (9% w/w), in Miglyol® 812.

**[0027]** FIG. **13**: Plasma concentration profile of formulation 11 containing 2.0 mg spray dried CR845.HCl (18% w/w), trehalose (60% w/w), sodium caprate (22% w/w), in Miglyol® 812.

**[0028]** FIG. **14**: Plasma concentration profile of formulation 12 containing 2.0 mg spray dried CR845.HCl (23% w/w), trehalose (77% w/w), suspended in 10% capric acid, Miglyol® 812.

## DETAILED DESCRIPTION OF THE INVENTION

**[0029]** In one embodiment, the oral formulation of the invention includes a therapeutic agent comprising a peptide, and at least one absorption enhancer, the absorption enhancer includes a medium chain fatty acid or a salt of a medium chain fatty acid, and a medium chain fatty acid glyceride, wherein the medium chain fatty acid or the salt of the medium chain fatty acid comprises capric acid or a salt of capric acid.

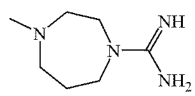
**[0030]** In another embodiment, the oral formulation of the invention includes a kappa opioid receptor agonist peptide, and at least one absorption enhancer, the absorption enhancer includes a medium chain fatty acid or a salt of a medium chain fatty acid, and a medium chain fatty acid glyceride, wherein the medium chain fatty acid or the salt of the medium chain fatty acid comprises capric acid or a salt of capric acid.

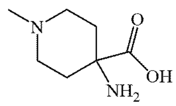
**[0031]** In another embodiment, the oral formulation of the invention includes a kappa opioid receptor agonist that includes one or more D-amino acids, and at least one absorption enhancer, the absorption enhancer includes a medium chain fatty acid or a salt of a medium chain fatty acid, and a medium chain fatty acid glyceride, wherein the medium chain fatty acid or the salt of the medium chain fatty acid comprises capric acid or a salt of capric acid. The salt of capric acid can be any suitable salt of capric acid, such as for instance, sodium caprate.

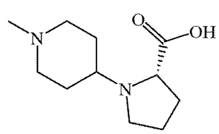
**[0032]** In another embodiment, the oral formulation of the invention includes a kappa opioid receptor agonist that includes one or more D-amino acids, and at least one absorption enhancer, the absorption enhancer includes a groups often exist in equilibrium with their zwitterionic forms. Thus, for any compound described herein that contains, for example, both amino and carboxyl groups, it will also be understood to include the corresponding zwitterion.

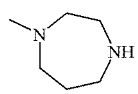
**[0062]** As used herein, the chemical designation "tetrapeptide-[ω(4-amino-piperidine-4-carboxylic acid)]" is used to indicate the aminoacyl moiety of the kappa opioid receptor agonist peptide amides of the invention derived from 4-aminopiperidine-4-carboxylic acid, wherein the nitrogen atom of the piperidine ring is bound to the C-terminal carbonyl-carbon of the tetrapeptide fragment, unless otherwise indicated.

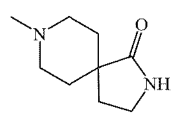
**[0063]** In another embodiment, the amide moiety of the D-amino acid amide (designated as "G" in structural formula I and formula III, below) is chosen from the following groups:

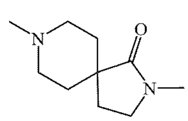


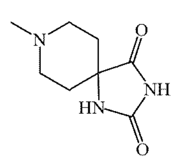


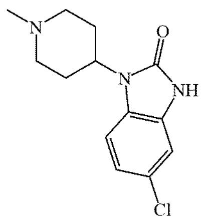


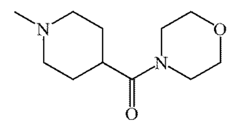


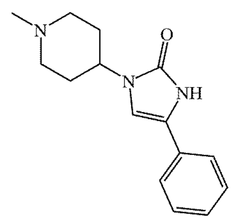


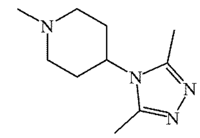


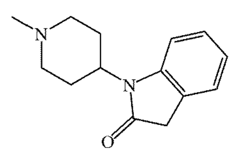


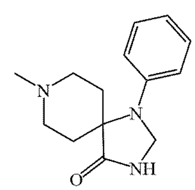


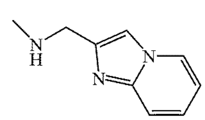


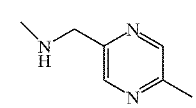


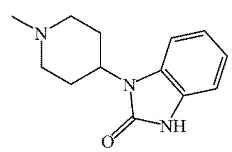


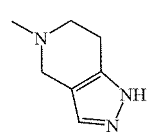


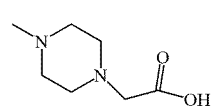


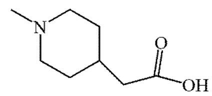


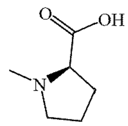


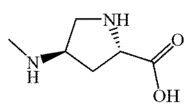


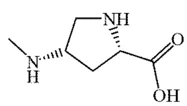


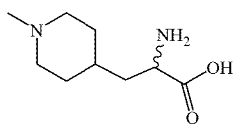


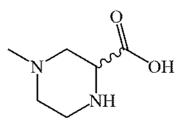


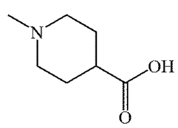


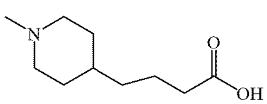


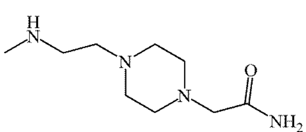


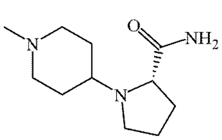


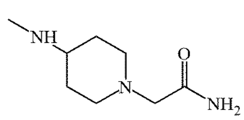


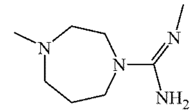


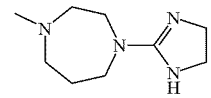


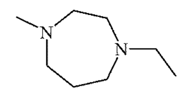


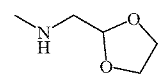


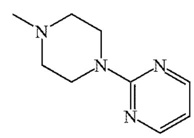


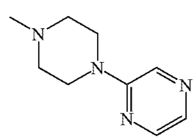


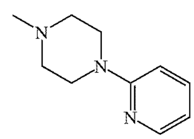


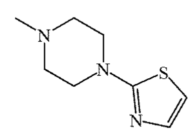


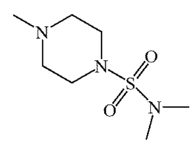


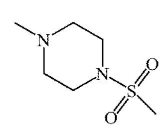


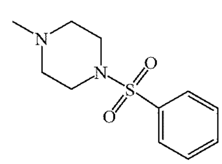












**[0140]** The bioactive composition of the invention including a biologically active peptide embedded in an oligomeric saccharide particle can be included in a pharmaceutically acceptable tablet, caplet, capsule, powder, or liquid suspension for administration as a medicament. The pharmaceutically acceptable tablet, caplet, capsule, powder, or liquid suspension may further include one or more of a salt of a carboxylic acid, an absorption enhancer, a binding agent and a pharmaceutically acceptable carrier or excipient. In one embodiment the pharmaceutically acceptable tablet, caplet, capsule, powder, or liquid suspension may include sodium citrate as the carboxylic acid. In another embodiment the pharmaceutically acceptable tablet, caplet, capsule, powder, or liquid suspension may include lauroyl L-carnitine as an absorption enhancer.

**[0141]** In a further embodiment, the pharmaceutically acceptable tablet, caplet, capsule, powder, or liquid suspension may include CR845 as the a kappa opioid receptor agonist biologically active peptide embedded in an oligomeric saccharide in the form of stabilized particles having a diameter of about 50 microns. In some embodiments, the oligomeric saccharide is a glucose-containing oligomeric saccharide, such as trehalose.

**[0142]** The formulations including the kappa opioid receptor agonist peptide amides and absorption enhancers of the invention can be administered by methods disclosed herein for the treatment or prevention of any hyperalgesic condition, such as, but without limitation, a hyperalgesic condition associated with allergic dermatitis, contact dermatitis, skin ulcers, inflammation, rashes, fungal irritation and hyperalgesic conditions associated with infectious agents, burns, abrasions, bruises, contusions, frostbite, rashes, acne, insect bites/stings, skin ulcers, mucositis, gingivitis, bronchitis, laryngitis, sore throat, shingles, fungal irritation, fever blisters, boils, Plantar's warts, surgical procedures or vaginal lesions.

**[0143]** Moreover, the formulations including the kappa opioid receptor agonist peptide amides and absorption enhancers of the invention can be administered by methods disclosed herein for the treatment or prevention of any hyperalgesic condition associated with burns, abrasions, bruises, abrasions (such as corneal abrasions), contusions, frostbite, rashes, acne, insect bites/stings, skin ulcers (for instance, diabetic ulcers or a decubitus ulcers), mucositis, inflammation, gingivitis, bronchitis, laryngitis, sore throat, shingles, fungal irritation (such as athlete's foot or jock itch), fever blisters, boils, Plantar's warts or vaginal lesions (such as vaginal lesions associated with mycosis or sexually transmitted diseases).

**[0144]** Hyperalgesic conditions associated with post-surgery recovery can also be addressed by administration of formulations including the kappa opioid receptor agonist peptide amides and absorption enhancers of the invention. The hyperalgesic conditions associated with post-surgery recovery can be any hyperalgesic conditions associated with post-surgery recovery, such as for instance, radial keratectomy, tooth extraction, lumpectomy, episiotomy, laparoscopy and arthroscopy. Hyperalgesic conditions associated with inflammation can also be addressed by administration of formulations including the kappa opioid receptor agonist peptide amides and absorption enhancers of the invention.

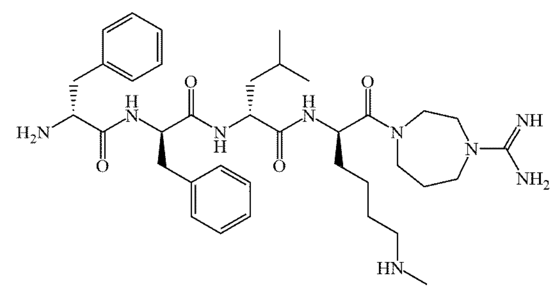
EXAMPLES

**[0145]** The following listing of D-amino acid peptide amides listed as compounds (1) - (103) can be used in the formulations of the present invention:

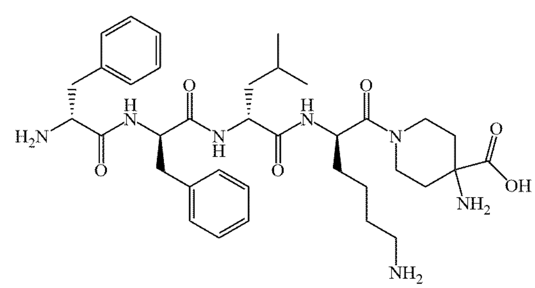
D-Amino Acid Peptide Amides

Examples (1) - (103):

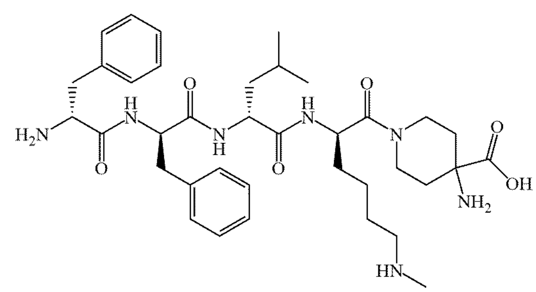
**[0146]** Compound (1): D-Phe-D-Phe-D-Leu-(ε-Me)D-Lys-[4-amidinohomopiperazine amide]:



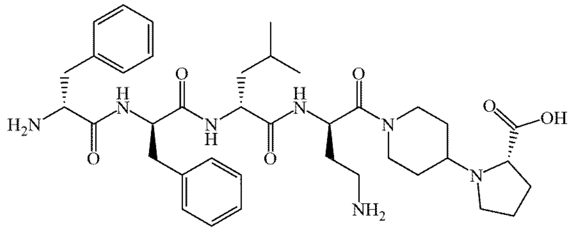
Compound (2): D-Phe-D-Phe-D-Leu-D-Lys-[ω(4-aminopiperidine-4-carboxylic acid)]-OH:



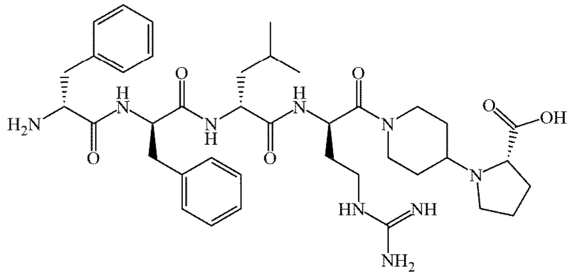
Compound (3): D-Phe-D-Phe-D-Leu-(ε-Me)D-Lys-[ω(4-aminopiperidine-4-carboxylic acid)]-OH:



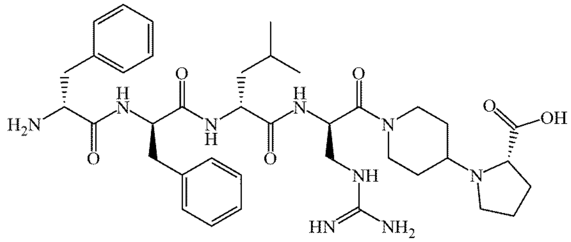
Compound (4): D-Phe-D-Phe-D-Leu-D-Lys-[N-(4-piperidinyl)-L-proline]-OH:



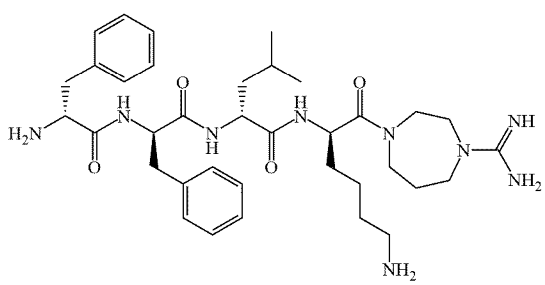
Compound (13): D-Phe-D-Phe-D-Leu-D-Nar-[N-(4-piperidinyl)-L-proline]-OH:



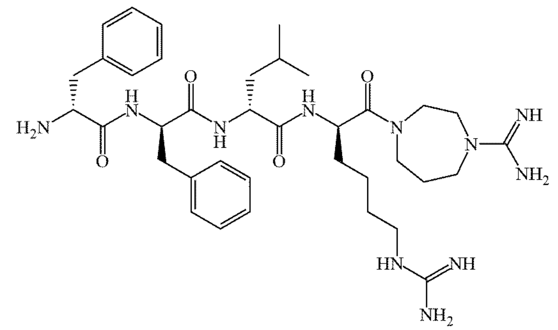
Compound (14): D-Phe-D-Phe-D-Leu-D-Dap(amidino)-[N-(4-piperidinyl)-L-proline]-OH:



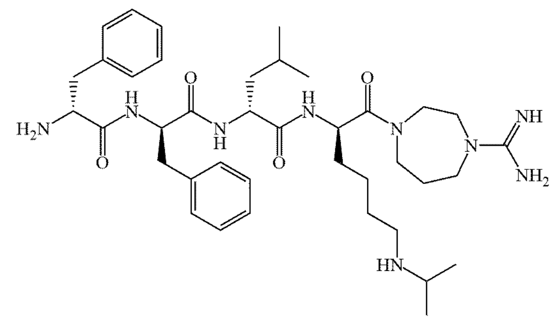
Compound (15): D-Phe-D-Phe-D-Leu-D-Lys-[4-amidinohomopiperazine amide]:



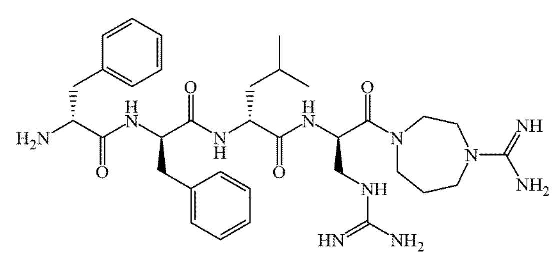
Compound (16): D-Phe-D-Phe-D-Leu-D-Har-[4-amidinohomopiperazine amide]:



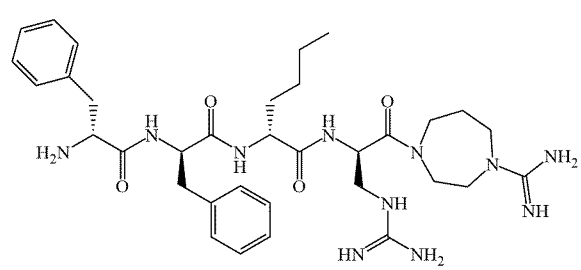
Compound (17): D-Phe-D-Phe-D-Leu-(ε-iPr)D-Lys-[4-amidinohomo-piperazine amide]:



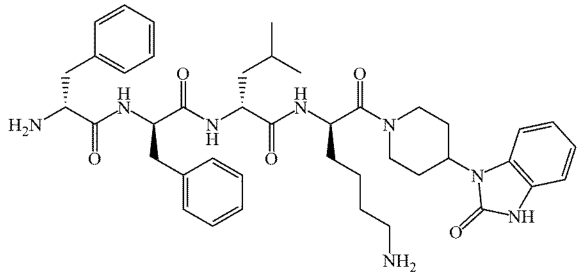
Compound (18): D-Phe-D-Phe-D-Leu-(β-amidino)D-Dap-[4-amidinohomopiperazine amide]:



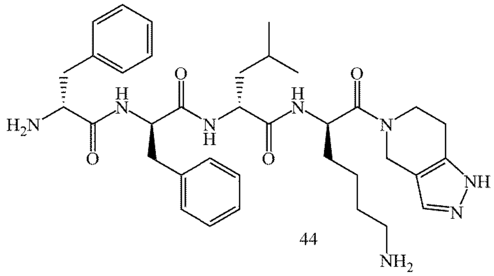
Compound (19): D-Phe-D-Phe-D-Nle-(β-amidino)D-Dap-[4-amidinohomopiperazine amide]:



Compound (20): D-Phe-D-Phe-D-Leu-(β-amidino)D-Dap-[homopiperazine amide]:

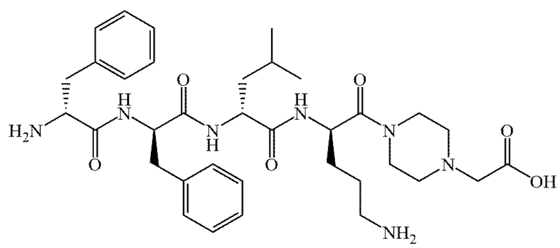


Compound (37): D-Phe-D-Phe-D-Leu-D-Lys-[4,5,6,7-tetrahydro-1H-pyrazolo[4,3-c]pyridine amide]:

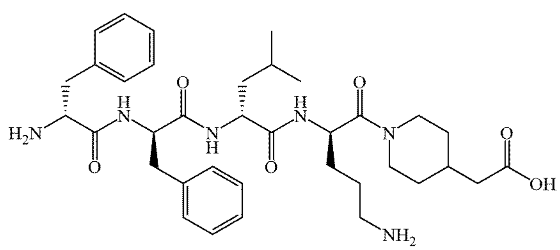


44

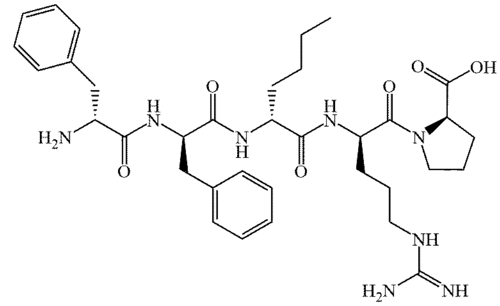
Compound (38): D-Phe-D-Phe-D-Leu-D-Orn-[4-(2-aminoethyl)-1-carboxymethyl-piperazine]-OH:



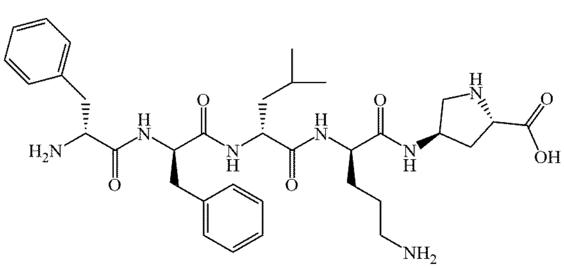
Compound (39) D-Phe-D-Phe-D-Leu-D-Orn-[4-carboxymethyl-piperidine]-OH:



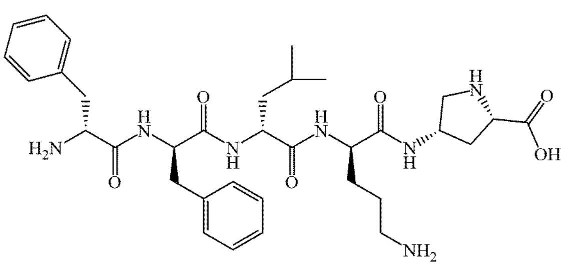
Compound (40) D-Phe-D-Phe-D-Nle-D-Arg-D-Pro-OH:



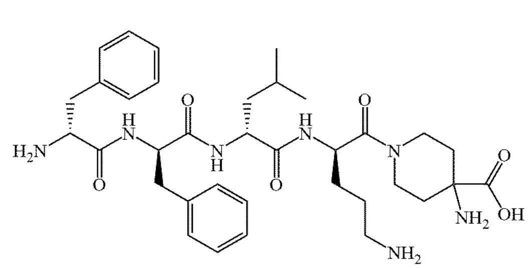
Compound (41) D-Phe-D-Phe-D-Leu-D-Orn-[(2S,4R)-4-amino-pyrrolidine-2-carboxylic acid]-OH:



Compound (42) D-Phe-D-Phe-D-Leu-D-Orn-[(2S,4S)-4-amino-pyrrolidine-2-carboxylic acid]-OH:

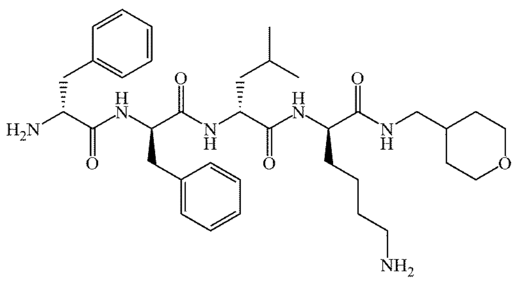


Compound (43) D-Phe-D-Phe-D-Leu-D-Orn-[ω(4-amino-piperidine-4-carboxylic acid)]-OH:

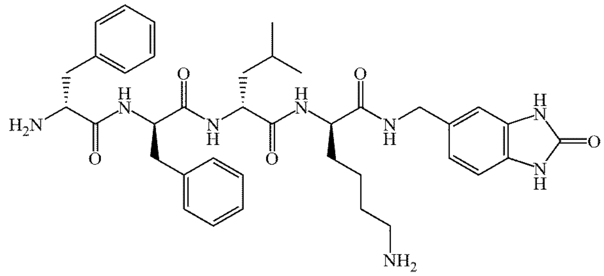


45

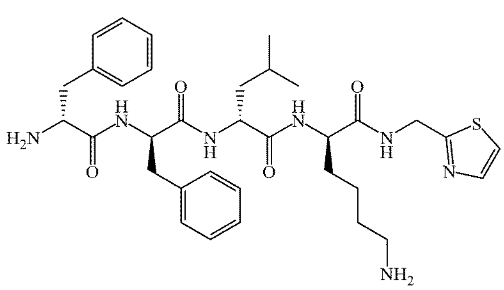
Compound (44) D-Phe-D-Phe-D-Leu-D-Orn-[ω(D/L-2-amino-3-(4-N-piperidinyl)propionic acid)]-OH:



Compound (74): D-Phe-D-Phe-D-Leu-D-Lys-[5-(amino-methyl)-1H-benzo[d]imidazol-2(3H)-one amide]:

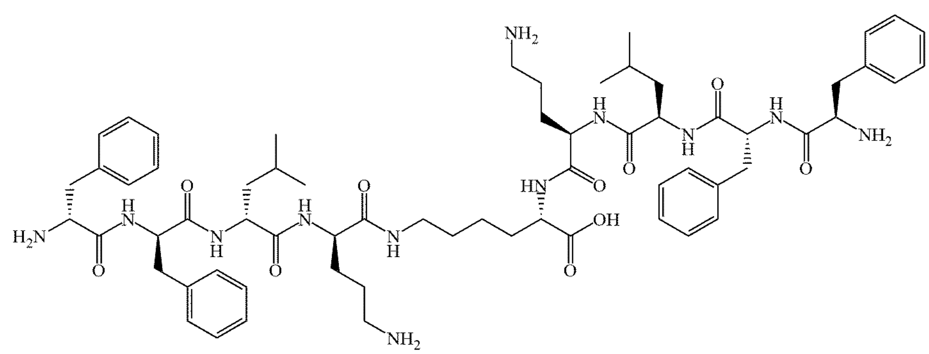


Compound (75): D-Phe-D-Phe-D-Leu-D-Lys-N-(thiazol-2-ylmethyl) amide:



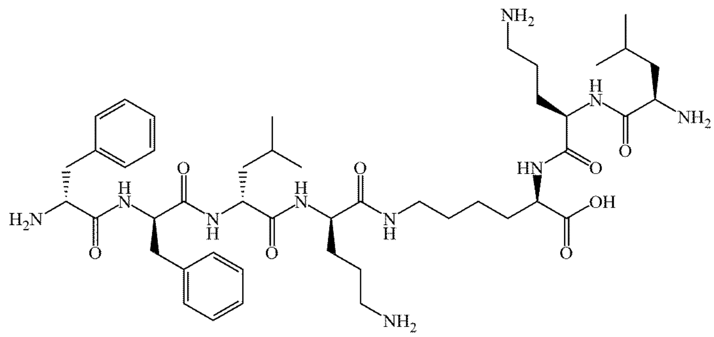
Compound (76):

D—Phe—D—Phe—D—Leu—D—Orn—εLys-OH
D—Phe—D—Phe—D—Leu—D—Orn



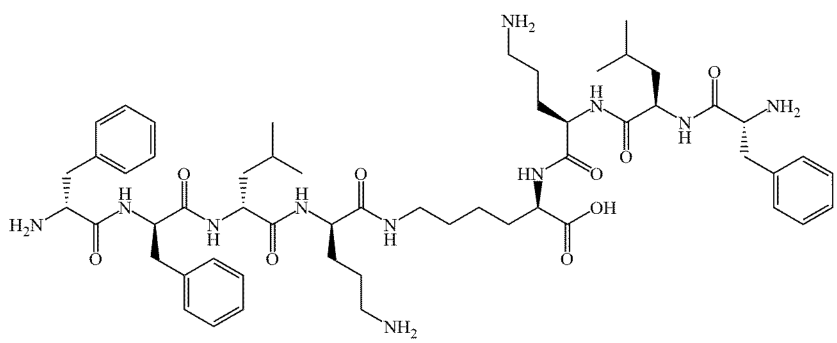
Compound (77):

D—Phe—D—Phe—D—Leu—D—Orn—εLys—OH
D—Leu—D—Orn



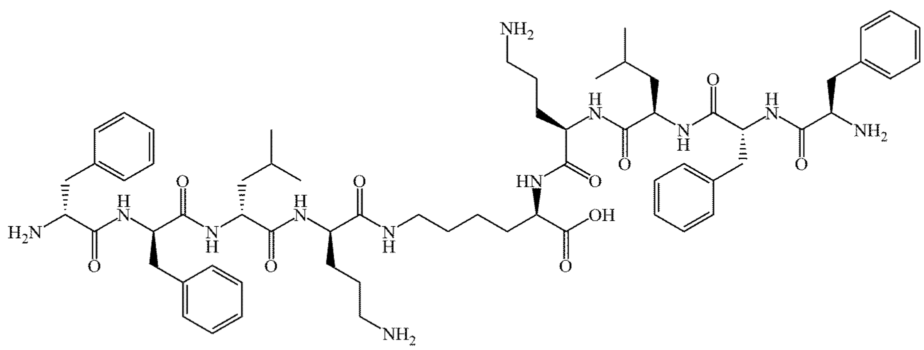
Compound (84):

D-Phe—D-Phe—D-Leu—D-Orn—εLys—OH
D-Phe-D-Leu-D-Orn

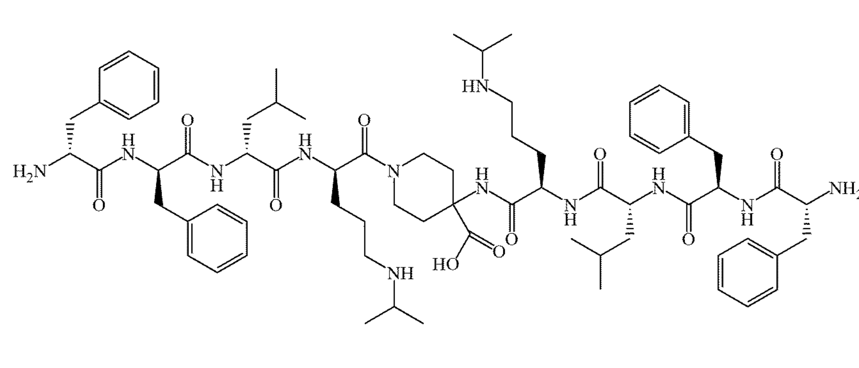


Compound (85):

D-Phe—D-Phe—D-Leu—D-Orn—εLys—OH
D-Phe-D-Phe-D-Leu-D-Orn



Compound (86): 1N,4N-bis-[D-Phe-D-Phe-D-Leu-(iPr)D-Orn]-4-amino-4-carboxylic-piperidine



**[0166]** FIGS. **10-14** show bioactivity profiles for formulations 8-12. Table 6 shows the bioactivity determined as described above for each of these formulations:

TABLE 6

| Formulation: Suspension Matrix | %f | Std. Dev. | SEM |
|---|---|---|---|
| 8. 10% Capric acid/Miglyol® | 16.3 | 14.92 | 5.27 |
| 9. Miglyol® | 11.5 | 6.07 | 2.15 |
| 10. Miglyol® | 12.4 | 8.95 | 3.17 |
| 11. Miglyol® | 18.4 | 11.64 | 4.11 |
| 12. 10% Capric acid/Miglyol® | 14.5 | 10.89 | 3.85 |

**[0167]** The specifications of each of the U.S. patents and published patent applications, and the texts of the literature references cited in this specification are herein incorporated by reference in their entireties. In the event that any definition or description contained found in one or more of these references is in conflict with the corresponding definition or description herein, then the definition or description disclosed herein is intended.

**[0168]** The examples provided herein are for illustration purposes only and are not intended to limit the scope of the invention, the full breadth of which will be readily recognized by those of skill in the art.

What is claimed is:

**1**. A formulation for oral delivery of a kappa opioid receptor agonist, the formulation comprising:

the kappa opioid receptor agonist and an absorption enhancer, wherein the absorption enhancer comprises:

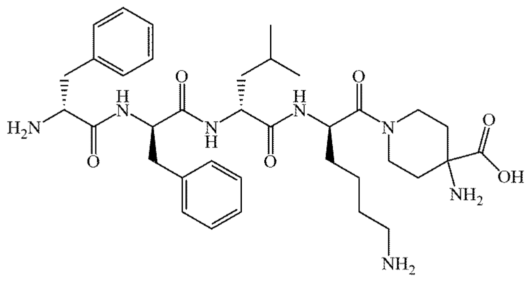
(a) a medium chain fatty acid or a salt of a medium chain fatty acid;

(b) a medium chain fatty acid glyceride; or

(c) (i) a medium chain fatty acid or a salt of a medium chain fatty acid and (ii) a medium chain fatty acid glyceride.

**2**. The formulation according to claim **1**, wherein kappa opioid receptor agonist is selected from the group consisting of:

(i) CR845 having the formula:



D-Phe-D-Phe-D-Leu-D-Lys-[ω(4-aminopiperidine-4-carboxylic acid)]-OH;

(ii) to asimadoline (N-[(1S)-2-[(3S)-3-hydroxy-pyrrolidin-1-yl]-1-phenylethyl]-N-methyl-2,2-diphenylacetamide), and

(iii) nalfurafine ((2E)-N-[(5α,6β)-17-(cyclopropylmethyl)-3,14-dihydroxy-4,5-epoxymorphinan- 6-yl]-3-(3-furyl)-N-methylacrylamide).

**3**. The formulation according to claim **1**, wherein the medium chain fatty acid comprises capric acid and/or a capric acid salt.

**4**. The formulation according to claim **1**, wherein the medium chain fatty acid glyceride comprises a medium chain fatty acid triglyceride.

**5**. The formulation according to claim **4**, wherein the medium chain triglyceride is selected from the group consisting of Miglyol® 810, Miglyol® 812, Capmul® MCM, Captex® 100, Captex® 200, Captex® 300, Captex® 355, Neobee® 1053, Neobee® M5, Capmul® PGMC, Tween® 60, Tween® 80, Lubrisol® ALF, Kolliphor® EL, Kolliphor® HS 15, and Gelucare® 44/14.

**6**. The formulation according to claim **2**, wherein the formulation further comprises a pharmaceutically acceptable diluent, excipient or carrier.

**7**. A capsule containing a formulation for oral delivery of a therapeutic agent, wherein the formulation comprises a kappa opioid receptor agonist and an absorption enhancer, wherein the absorption enhancer comprises: a medium chain fatty acid or a salt of a medium chain fatty acid; and a medium chain fatty acid glyceride, wherein the capsule is an enteric coated capsule or a capsule having intrinsic enteric properties.

**8**. A bioactive composition comprising a kappa opioid receptor agonist embedded in an oligomeric saccharide comprising one or more glucose monomers forming a particle comprising a stabilized kappa opioid receptor agonist.

**9**. The bioactive composition according to claim **8**, wherein the kappa opioid receptor agonist comprises CR845 and the oligomeric saccharide comprises trehalose.

**10**. The bioactive composition according to claim **9**, further comprising one or more of a salt of a carboxylic acid, an absorption enhancer, a binding agent, a chelating agent and a pharmaceutically acceptable carrier or excipient.

**11**. The bioactive composition according to claim **10**, wherein the salt of the carboxylic acid comprises sodium citrate, the chelating agent comprises EDTA and the absorption enhancer comprises lauroyl L-carnitine.

**12**. The bioactive composition according to claim **9**, further comprising one or more medium chain fatty acid or one or more salt of a medium chain fatty acids, and a medium chain fatty acid glyceride.

**13**. A pharmaceutically acceptable tablet, caplet, capsule, powder, or liquid suspension comprising the bioactive composition according to claim **9**.

**14**. The bioactive composition according to claim **13**, wherein the liquid suspension further comprises one or more of an absorption enhancer, and a pharmaceutically acceptable ($C_8$-$C_{12}$) short chain fatty acid triglyceride.

**15**. The bioactive composition according to claim **14**, further comprising one or more of an emulsifying agent, a salt of a carboxylic acid, an absorption enhancer, a binding agent and a pharmaceutically acceptable carrier or excipient.

* * * * *